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# main.py

from fastapi import FastAPI, Form
from fastapi.responses import HTMLResponse
from pydantic import BaseModel
import sqlite3
from datetime import datetime, timedelta

app = FastAPI()

# DATABASE SETUP
conn = sqlite3.connect("study_planner.db", check_same_thread=False)
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS study_plans (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    subject TEXT,
    total_topics INTEGER,
    days INTEGER,
    hours_per_day INTEGER,
    created_at TEXT
)
""")

conn.commit()

# MODEL
class StudyPlan(BaseModel):
    subject: str
    total_topics: int
    days: int
    hours_per_day: int

# HOME PAGE
@app.get("/", response_class=HTMLResponse)
def home():
    return """
<html>
<head>
    <title>AI Study Planner</title>
    <style>

```

```

body {
  font-family: Arial;
  background-color: #f4f4f4;
  padding: 40px;
}

.container {
  background: white;
  padding: 30px;
  border-radius: 10px;
  width: 400px;
  margin: auto;
  box-shadow: 0px 0px 10px rgba(0,0,0,0.1);
}

input {
  width: 100%;
  padding: 10px;
  margin-top: 10px;
  margin-bottom: 20px;
}

button {
  background: #007bff;
  color: white;
  border: none;
  padding: 12px;
  width: 100%;
  border-radius: 5px;
  cursor: pointer;
}

h1 {
  text-align: center;
}
</style>
</head>

<body>
  <div class="container">
    <h1>AI Study Planner</h1>

    <form action="/generate-plan" method="post">

```

```

<label>Subject</label>
<input type="text" name="subject" required>

<label>Total Topics</label>
<input type="number" name="total_topics" required>

<label>Days Available</label>
<input type="number" name="days" required>

<label>Study Hours Per Day</label>
<input type="number" name="hours_per_day" required>

<button type="submit">Generate Plan</button>
</form>
</div>
</body>
</html>
"""

```

GENERATE STUDY PLAN

@app.post("/generate-plan", response_class=HTMLResponse)

```

def generate_plan(
    subject: str = Form(...),
    total_topics: int = Form(...),
    days: int = Form(...),
    hours_per_day: int = Form(...)
):

```

Save to database

```

cursor.execute("""
INSERT INTO study_plans
(subject, total_topics, days, hours_per_day, created_at)
VALUES (?, ?, ?, ?, ?)
""", (
    subject,
    total_topics,
    days,
    hours_per_day,
    str(datetime.now())
))

```

```

conn.commit()

```

```

topics_per_day = max(1, total_topics // days)

plan_html = ""

current_topic = 1

for day in range(1, days + 1):

    start_topic = current_topic
    end_topic = min(current_topic + topics_per_day - 1, total_topics)

    plan_html += f"""
    <tr>
        <td>Day {day}</td>
        <td>Topic {start_topic} - Topic {end_topic}</td>
        <td>{hours_per_day} Hours</td>
    </tr>
    """

    current_topic = end_topic + 1

    if current_topic > total_topics:
        break

return f"""
<html>
<head>
    <title>Study Plan</title>

    <style>
        body {{
            font-family: Arial;
            background: #f4f4f4;
            padding: 40px;
        }}

        .container {{
            background: white;
            padding: 30px;
            border-radius: 10px;
            max-width: 700px;
            margin: auto;
            box-shadow: 0px 0px 10px rgba(0,0,0,0.1);
        }}
    </style>

```

```

table {{
  width: 100%;
  border-collapse: collapse;
  margin-top: 20px;
}}

th, td {{
  border: 1px solid #ddd;
  padding: 12px;
  text-align: center;
}}

th {{
  background: #007bff;
  color: white;
}}

h1 {{
  text-align: center;
}}
</style>
</head>

<body>

  <div class="container">

    <h1>{subject} Study Plan</h1>

    <table>
      <tr>
        <th>Day</th>
        <th>Topics</th>
        <th>Study Time</th>
      </tr>

      {plan_html}

    </table>

  </div>

</body>

```

</html>
""""

```
# RUN SERVER
# command:
# uvicorn main:app --reload
```